Kartikey Pathak

+91 7506048765 | krpathak@proton.me | LinkedIn | GitHub | Portfolio Website

Master resume — full record. Cut down per application.

Education

Veermata Jijabai Technological Institute (VJTI) — Mumbai, India
B.Tech in Electrical Engineering, Minor in Robotics (CGPA: 8.04) — *2023 – 2027*

Matoshree Prabodhini Jr. College — Maharashtra, India
HSC, Maharashtra State Board — MHT-CET 99.14 percentile — *2023*

Experience

Eyecandy Robotics — April 2026 – August 2026
Electronics and Hardware Engineer — *Hybrid*

- Designed production PCBs for a legged consumer robot: Raspberry Pi Zero 2W and Milk-V compute carriers, 1S/2S battery charging and power boards (BQ25887, MP2672, TPS562202 rails), a servo driver board, a microphone board and inter-board connector boards
- Owned schematic capture, routing, DFM review and fabrication output for each board
- Integrated MPU6050 IMU, a buffered Feetech servo bus, per-leg force-sensitive-resistor conditioning (LM393 comparators), PAM8403 audio and a ReSpeaker microphone array at board level
- Laid out test points and solder-jumper configuration options so each subsystem could be probed or bypassed independently during hardware bring-up
- Wrote fleet-provisioning tooling that clones one built Raspberry Pi runtime image onto many robots with unique machine IDs, hostnames and SSH host keys, so a fleet coexists on one network without identity or IP collisions

NishCorp Technologies — July 2025 – November 2025
Robotics Control Systems Engineer Intern — *Hybrid*

- Developed Gazebo simulation for a stair-climbing robotic wheelchair platform to analyze motion and steering behaviour, converting CAD to URDF and bringing the model up in Gazebo and RViz
- Applied differential drive kinematic models, integrated an IMU plugin, and wrote teleoperation so the platform could be driven and studied in simulation before it was built in steel
- Contributed to the half-scale hardware prototype through 3D printing, assembly, motor electronics, integration, testing and field evaluation; the prototype could sit and stand under its own power
- Simulation work formed part of the technical case that helped the startup raise Rs 13 lakh in grants

Krishna Defence and Allied Industries Limited — April 2025 – July 2025
Research Intern - Defence Technology — *IIT Gandhinagar*

- Solved indoor localization for an autonomous forklift in a naval dockyard, where GPS is blocked by surrounding steel and wheel odometry drifts on every slip
- Surveyed Bluetooth, LoRa, RFID, radar and camera-based options before selecting ultra-wideband ranging
- Developed firmware for Qorvo DWM3001CDK Ultra-Wideband modules using Zephyr RTOS, handling low-level drivers and RF packets
- Implemented a custom double-sided two-way ranging protocol, tuning channel selection, preamble length and PAC size and re-deriving timing delays per configuration, extending usable range from the stock 45 m to 315 m
- Implemented SLAM and NAV2 on a Raspberry Pi 4B fusing UWB, IMU and LIDAR for autonomous forklift POC navigation in warehouse environments
- Scaled testing from a 2×2 m room to a 50×60 m yard and validated against a surveyor's total station rather than internal estimates: localization error of 30 cm across a 10,000 sq meter area and 2 cm in a 50 sq meter area

Projects

PCB Axial-Flux BLDC Motor | *KiCAD, FEMM, Python* — Aug 2026 – Present

- Designing a coreless axial-flux motor whose stator is the PCB itself: 100 mm two-layer board, 6 wedge coils over 8 poles, dual 3D-printed rotor carrying 16 N42 magnets across a 0.5 mm air gap
- Generate the coil geometry programmatically rather than drawing it, so pole count, trace width and radii are parameters of the design
- Established that a top-down 2D cut cannot yield torque for an axial-flux machine, and built the magnetostatic model by unrolling the machine on a cylinder into five radial slices so that coil current runs out of plane
- Extracted torque constant, back-EMF, Kv and torque ripple from the solved air-gap flux, predicting 40.9 mN·m/A rms, 496 rpm/V and zero cogging torque for the coreless design
- Identified before fabrication that the 3D-printed rotor gives away 3.3x the torque of the steel-backed design the board was sized for, and that the motor is thermally rather than magnetically limited; sized 6 mm steel back plates as the fix by checking plate saturation

FOC Controller for Micro BLDC Motor | *STM32, DRV8311H, PCB Design* — Jan 2026 – Feb 2026

- Developed a Field-Oriented Control (FOC) system for a 4300 KV micro BLDC motor operating from a 3.7 V supply
- Designed and routed a custom PCB integrating an STM32 microcontroller, DRV8311H motor driver, and AS5600 encoder
- Implemented centre-aligned three-phase PWM at 20 kHz with a 40 kHz control ISR, zero-sequence injection, AS5600 angle over I2C and DMA-fed phase current sensing on an STM32G431

Smart Switch Board | *ESP32, ESP-IDF, FreeRTOS, MQTT, BLE, Flutter, PCB Design* 2025 – Present

- Built a network-controlled mains wall switch across every layer alone: CAD enclosure, a perf board carrying relays and optocouplers to isolate 220 V from a 3.3 V ESP32, ESP-IDF firmware, a TLS MQTT broker and a Flutter application
- Held to a hard requirement that the physical switches keep working with network, broker and app all unavailable
- Implemented WiFi provisioning over BLE with no hardcoded credentials, switch state persisted in NVS across power cuts, and a 5-second heartbeat so switch, firmware, broker and app stay agreed on state
- Installed and running at home; next steps are a routed KiCAD PCB in place of the perf board and OTA firmware updates

MARIO — 3DOF Robotic Manipulator | *ROS2, Gazebo Fortress, ESP32, micro-ROS* Apr 2025 – Aug 2025

- Developed embedded firmware in ESP-IDF and a micro-ROS communication pipeline for real-time hardware-software integration, so one command set drives both the simulation and the physical arm
- Migrated the simulation environment off end-of-life Gazebo Classic to Gazebo Fortress, porting model, plugins and launch files, and rewrote teleoperation for beginners
- Manufactured, tested and distributed 50+ robotics kits used by 200+ students for learning manipulator control

Kurma - Quadruped Robot | *ESP-IDF, MQTT, Kinematics* Jan 2025 – Mar 2025

- Designed and built a turtle-inspired quadruped robot, developing multiple gait patterns for stable multi-terrain locomotion
- Developed a teleoperation system utilizing MQTT for low-latency communication between a Raspberry Pi and ESP32s
- Used a statically stable gait with three feet always grounded, which kept the robot walking on rough ground that stops wheeled platforms

Voice Video Manipulator | *ROS2, Gazebo, RViz, Computer Vision* June 2024 – July 2024

- Programmed forward and inverse kinematics for an Open Manipulator X robot arm to perform precise pick-and-place tasks
- Implemented object detection and tracking algorithms to locate target objects in the Gazebo simulation environment
- Chained speech recognition to object detection to inverse kinematics, so a spoken target became an arm trajectory; first project built with ROS2 nodes and topics rather than a single control loop

Automated Coin Sorter | *Raspberry Pi, TensorFlow, OpenCV, Servos* 24-Hour Hackathon

- Classified Rs 1, 2, 5 and 10 coins from a camera feed using a CNN trained by transfer learning, reaching above 95% accuracy on Rs 5 and Rs 10 coins
- Routed each classified coin into its bin using servo-actuated gates driven from the Raspberry Pi

Additional Projects

Self-Balancing Robot (Wall-E) | *ESP32, ESP-IDF, MPU6050, PID* 2024

- Balanced a two-wheeled robot by fusing gyroscope and accelerometer with a complementary filter and running a PID loop at a few hundred hertz; the platform SRA now teaches sensor fusion on

Maze Solving Robot | *ESP32, ESP-IDF, IR Sensor Array* 2024

- Mapped a maze with the left-hand rule at a 100 Hz control loop, recorded junctions, then replayed the shortest path; placed 4th in the SRA maze competition

Gesture Controlled Car | *ESP32, MPU6050, ESP-NOW* Dec 2023 – Jan 2024

- Drove a car by hand tilt, reading a glove-mounted IMU on one ESP32 and transmitting over ESP-NOW to a second ESP32 running the motor drivers; built during a first-semester winter internship

Automatic Animal Feeder | *Servos, 3D Printing, Timer Control* 2024

- Built a scheduled dispenser that feeds the robotics lab cat through a 3D-printed chute when the lab is empty; still running, with an open-source release planned for animal shelters

River Cleaning Boat | *RC Control, Mechanical Design, Waterproofing* 2024

- Built a remote-controlled boat with a front scoop and internal basket to collect floating trash, for an environmental studies course; the engineering problem was waterproofing electronics on a student budget

Competitions

eYantra Warehouse Drone Competition | *IIT Bombay* Oct 2024 – Feb 2025

- Developed autonomous navigation and PID control algorithms for stable hovering and motion of a simulated drone
- Implemented ArUco marker detection using computer vision for autonomous localization and task execution
- Reached the national semifinals; the fragility of camera-only localization here led directly to the UWB work at Krishna Defence

Mass Robotics Challenge | *Quadruped Locomotion* 2025

- Entered Kurma, the turtle-inspired quadruped, advancing through to later competition rounds

Positions of Responsibility

Society of Robotics and Automation (SRA) Aug 2024 – Mar 2026
Mechanical Head & Active Member *VJTI, Mumbai*

- Conducted technical workshops on Embedded C, ESP32 microcontrollers, and ROS-based systems for 200+ students
- Mentored junior teams in developing self-balancing and line-following robots, providing hardware/software expertise
- Taught sensor fusion in practice, covering MPU6050 integration and complementary filtering for orientation estimation

Pratibimb, VJTI Cultural Festival Aug 2023 – Mar 2025
Department Head, Electrical Engineering *VJTI, Mumbai*

- Led the Electrical Engineering branch contingent across events, coordinating teams and participation

Achievements

- Led the Electrical Engineering branch to its first overall win at Pratibimb in the competition's 26-year history
- Directed *Plot Se Panga*, a short film that won first prize across all branches at VJTI
- Reached the national semifinals of the eYantra Warehouse Drone Competition, IIT Bombay
- Placed 4th in the SRA maze-solving robot competition
- Consistent team performances across hackathons through first and second year at VJTI

Technical Skills

Programming Languages: C, C++, Python, Embedded C
Software & Frameworks: ROS2, Gazebo, RViz, ESP-IDF, Zephyr RTOS, STM32CubeIDE, KiCAD, micro-ROS
Hardware: ESP32, STM32, Raspberry Pi, UWB Modules, Motor Drivers, BLDC/Stepper/DC Motors, LiDAR
Core Competencies: Robot Kinematics, Field-Oriented Control, SLAM, NAV2, PID Control, PCB Design, IoT